

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	R.DINESH	Reg. No.:	192524130
Section / Batch:	2ND Year AIDS	Date:	22/07/2026

Code of Conduct

I R.DINESH (192524130) (Name/ Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R.DINESH

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculatio

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____


SIMATS Online Programming Lab
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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

Q1 Binary Search 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;
    while (low < high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int binarySearch(int arr[], int n, int key) {
3     int low = 0, high = n - 1;
4     while (low < high) {
5         int mid = (low + high) / 2;
6         if (arr[mid] == key)
7             return mid;
8         else if (arr[mid] < key)
9             low = mid + 1;
10        else
11            high = mid - 1;
12    }
13    return -1;
14 }
15 int main() {
16     int arr[] = {10, 20, 30, 40, 50};
17     int n = 5;
18     int key = 30;
19     int result = binarySearch(arr, n, key);
20     printf("%d\n", result);
21     return 0;
22 }
```

Submitted

Test Input

5
10 20 30 40 50
30

Output

3

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init()
7 {
8     top = -1;
9 }
10
11 int main()
12 {
13     init();
14     printf("top=%d\n", top);
15     return 0;
16 }
```

Submitted

Test Input

stdin input (optional)

Output

top=-1

Pending manual review

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5];
4 int top;
5
6 void init()
7 {
8     top = -1;
9 }
10
11 int main()
12 {
13     init();
14     printf("top=%d\n", top);
15     return 0;
16 }
```

 Submitted

Test Input

stdin input (optional)

Output

top=-1

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
#(ch == '\n') {  
while(stack[top] != '\n') {  
    printf("%c", stack[top]);  
    pop();  
}  
pop();  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[5];  
4 int top = -1;  
5  
6 int main()  
7 {  
8     char ch = '(';  
9     stack[++top] = '(';  
10  
11     if (ch == '\n')  
12     {  
13         while (stack[top] != '\n')  
14             top--;  
15  
16         top--;  
17     }  
18  
19     printf("done");  
20     return 0;  
21 }
```

Submitted

Test Input

stdin input (optional)



Output

done

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structure
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q6 Stack

5 marks

In balancing Bracket, you identify the error.

```
if(ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[') {
    pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[5], top = -1;
4
5 int main()
6 {
7     char ch = '[';
8
9     stack[++top] = '[';
10
11     if ((ch == '[' && stack[top] == '[') ||
12         (ch == ']' && stack[top] == '['))
13     {
14         top--;
15     }
16     else
17     {
18         printf("unbalanced");
19         return 0;
20     }
21
22     printf("balanced");
23     return 0;
24 }
```

Submitted

Test Input

stdin input (optional)

Output

balanced

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structure

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

Q6 Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 int main()
10 {
11     struct node *head = malloc(sizeof(struct node));
12
13     head->data = 10;
14     head->next = NULL;
15
16     struct node *temp = head;
17
18     while (temp != NULL) {
19         printf("%d\n", temp->data);
20         temp = temp->next;
21     }
22
23     free(head);
24     return 0;
25 }
```

Submitted

Test Input

stdin input (optional)

Output

10

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List Data Structure

5 marks

Q9

Linked List

5 marks

Q10

Queue

5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack):
if top == 0:
    print("Underflow")
else:
    return stack[top--]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int stack[5] = {10}, top = 0;
6
7     if (top == -1)
8         printf("Underflow");
9     else
10         printf("%d", stack[top--]);
11
12     return 0;
13 }
```

Submitted

Test Input

stdin input (optional)

Output

10

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code
while current != NULL:
current->next = prev
prev = current
current = current->next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 struct node {
4     int data;
5     struct node *next;
6 };
7
8 int main()
9 {
10     struct node a = {10, NULL};
11     struct node b = {20, a};
12     struct node c = {30, b};
13
14     struct node *prev = NULL, *current = a, *next;
15
16     while (current != NULL) {
17         next = current->next;
18         current->next = prev;
19         prev = current;
20         current = next;
21     }
22
23     printf("%d -> %d -> %d", prev->data, prev->next->data, prev->next->next->data);
24
25     return 0;
26 }
```

Test Input

stdin input (optional)

Output

10 -> 20 -> 30

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1 ✓

Binary Search
5 marks

Q2 ✓

Stack
5 marks

Q3 ✓

Stack
5 marks

Q4 ✓

Stack
5 marks

Q5 ✓

Stack
5 marks

Q6 ✓

Linked List
5 marks

Q7 ✓

Stack
5 marks

Q8 ✓

Linked List Data Structures
5 marks

Q9 ✓

Linked List
5 marks

Q10 ✓

Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int main()  
4 {  
5     int head = 10;  
6     int new = 20;  
7  
8     printf("td -> td -> NULL", new, head);  
9  
10    return 0;  
11 }
```

Test Input

stdin input (optional)

Output

20 -> 10 -> NULL

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List Data Structures
5 marks**Q9**
Linked List
5 marks**Q10**
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int main()  
4 {  
5     int q[] = {10, 20, 30};  
6     int front = 0;  
7  
8     if (front == -1)  
9         printf("Underflow");  
10    else {  
11        printf("%d", q[front]);  
12        front++;  
13    }  
14  
15    return 0;  
16 }
```

Test Input

stdin input (optional)

Output

10

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Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
  
    else {  
        int op1 = pop();  
        int op2 = pop();  
  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3  
4 #define MAX 100  
5  
6 int stack[MAX];  
7 int top = -1;  
8  
9 void push(int x) {  
10     stack[++top] = x;  
11 }  
12  
13 int pop() {  
14     return stack[top--];  
15 }  
16  
17 int evaluator(char expression[]) {  
18     int i = 0;  
19  
20     while (expression[i] != '\0') {  
21         if (isdigit(expression[i])) {  
22             push(expression[i] - '0');  
23         } else {  
24             int op1 = pop();  
25             int op2 = pop();  
26  
27             switch (expression[i]) {  
28                 case '+':  
29                     push(op2 + op1);  
30                     break;  
31  
32                 case '-':  
33                     push(op2 - op1);  
34                     break;  
35  
36                 case '*':  
37                     push(op2 * op1);  
38                     break;  
39  
40                 case '/':  
41                     push(op2 / op1);  
42                     break;  
43             }  
44             i++;  
45         }  
46     }  
47  
48     return pop();  
49 }  
50  
51 int main() {  
52     char expression[] = "11+54*+9-";  
53  
54     printf("Result = %d\n", evaluator(expression));  
55  
56     return 0;  
57 }
```

Submitted

Test Input

stdin input (optional)

Output

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1, rear = -1;  
7  
8 void enqueue(int data)  
9 {  
10     if ((rear + 1) % MAX == front)  
11     {  
12         printf("Queue Overflow\n");  
13         return;  
14     }  
15  
16     if (front == -1)  
17     {  
18         front = rear = 0;  
19     }  
20     else  
21     {  
22         rear = (rear + 1) % MAX;  
23     }  
24  
25     queue[rear] = data;  
26 }  
27  
28 int dequeue()  
29 {  
30     if (front == -1)  
31     {  
32         printf("Queue underflow\n");  
33         return -1;  
34     }  
35  
36     int data = queue[front];  
37  
38     if (front == rear)  
39     {  
40         front = rear = -1;  
41     }  
42     else  
43     {  
44         front = (front + 1) % MAX;  
45     }  
46  
47     return data;  
48 }  
49  
50 void display()  
51 {  
52     if (front == -1)  
53     {  
54         printf("Queue is empty\n");  
55         return;  
56     }  
57  
58     int i = front;  
59  
60     printf("Queue: ");  
61  
62     while (1)  
63     {  
64         printf("%d ", queue[i]);  
65  
66         if (i == rear)  
67             break;  
68  
69         i = (i + 1) % MAX;  
70     }  
71  
72     printf("\n");  
73 }  
74  
75 int main()
```

Test Input

stdin input (optional)

Output

```
Queue: 10 20 30 40  
deleted element = 10  
deleted element = 20  
Queue: 30 40  
Queue: 30 40 50 60
```

Pending manual review

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notation - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[], int l, int r, int x)
4 {
5     if (l > r)
6         return -1;
7
8     int mid = l + (r - l) / 2;
9
10    if (arr[mid] == x)
11        return mid;
12
13    if (arr[mid] > x)
14        return binarySearch(arr, l, mid - 1, x);
15
16    return binarySearch(arr, mid + 1, r, x);
17 }
18
19 int main()
20 {
21     int arr[] = {0, 5, 9, 12, 14, 23, 39, 56, 72, 91};
22     int n = sizeof(arr) / sizeof(arr[0]);
23     int x = 23;
24
25     int result = binarySearch(arr, 0, n - 1, x);
26
27     if (result == -1)
28         printf("element not found\n");
29     else
30         printf("element found at index %d\n", result);
31
32     return 0;
33 }
```

Test Input

stdin input (optional)

Output

element found at index 5

Pending manual review

Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notation - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q4 Queue

5 marks

Find the error in this Deque insertfront operation.

```
void insertfront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int deque[MAX];  
6 int front = -1, rear = -1;  
7  
8 void insertfront(int data)  
9 {  
10     // Check Overflow  
11     if ((front == 0 && rear == MAX - 1) || (front == rear + 1))  
12     {  
13         printf("Queue Overflow\n");  
14         return;  
15     }  
16  
17     // insert element  
18     if (front == -1)  
19     {  
20         front = rear = 0;  
21     }  
22     else if (front == 0)  
23     {  
24         front = MAX - 1;  
25     }  
26     else  
27     {  
28         front = front - 1;  
29     }  
30  
31     deque[front] = data;  
32 }  
33  
34 void display()  
35 {  
36     if (front == -1)  
37     {  
38         printf("Queue is empty\n");  
39         return;  
40     }  
41  
42     int i = front;  
43  
44     while (1)  
45     {  
46         printf("%d ", deque[i]);  
47  
48         if (i == rear)  
49             break;  
50  
51         i = (i + 1) % MAX;  
52     }  
53  
54     printf("\n");  
55 }  
56  
57 int main()  
58 {  
59     insertfront(10);  
60     insertfront(20);  
61     insertfront(30);  
62  
63     display();  
64  
65     return 0;  
66 }
```

Submitted

Test Input

stdin input (optional)

Output

30 20 10

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Recursion
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int queue[MAX];  
6 int front = -1;  
7 int rear = -1;  
8  
9 void enqueue(int data)  
10 {  
11     if (rear == MAX - 1)  
12     {  
13         printf("Queue Overflow\n");  
14         return;  
15     }  
16  
17     if (front == -1)  
18     {  
19         front = 0;  
20         rear = 0;  
21     }  
22     else  
23     {  
24         rear++;  
25     }  
26  
27     queue[rear] = data;  
28 }  
29  
30 int dequeue()  
31 {  
32     if (front == -1 || front > rear)  
33     {  
34         printf("Queue underflow\n");  
35         return -1;  
36     }  
37  
38     int data = queue[front++];  
39  
40     if (front > rear)  
41     {  
42         front = -1;  
43         rear = -1;  
44     }  
45  
46     return data;  
47 }  
48  
49 void display()  
50 {  
51     if (front == -1 || front > rear)  
52     {  
53         printf("Queue is empty\n");  
54         return;  
55     }  
56  
57     for (int i = front; i <= rear; i++)  
58     {  
59         printf("%d ", queue[i]);  
60     }  
61  
62     printf("\n");  
63 }  
64  
65 int main()  
66 {  
67     enqueue(10);  
68     enqueue(20);  
69     enqueue(30);  
70  
71     display();  
72  
73     printf("deleted: %d\n", dequeue());  
74  
75     display();  
76  
77     return 0;  
78 }
```

Test Input

stdin input (optional)

Output

```
10 20 30  
deleted: 10  
20 30
```

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notation - Recursive Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct node  
5 {  
6     int data;  
7     struct node *prev;  
8     struct node *next;  
9 };  
10  
11 typedef struct node node;  
12  
13 void deletenode(node **head, node *del)  
14 {  
15     if (*head == NULL || del == NULL)  
16         return;  
17  
18     // if deleting the head node  
19     if (*head == del)  
20     {  
21         *head = del->next;  
22  
23         if (*head != NULL)  
24             (*head)->prev = NULL;  
25     }  
26  
27     // update next pointer of previous node  
28     if (del->prev != NULL)  
29         del->prev->next = del->next;  
30  
31     // update previous pointer of next node  
32     if (del->next != NULL)  
33         del->next->prev = del->prev;  
34  
35     free(del);  
36 }  
37  
38 void printlist(node *head)  
39 {  
40     while (head != NULL)  
41     {  
42         printf("%d ", head->data);  
43         head = head->next;  
44     }  
45  
46     printf("\n");  
47 }  
48  
49 int main()  
50 {  
51     node *head = (node *)malloc(sizeof(node));  
52     node *second = (node *)malloc(sizeof(node));  
53     node *third = (node *)malloc(sizeof(node));  
54  
55     head->data = 10;  
56     head->prev = NULL;  
57     head->next = second;  
58  
59     second->data = 20;  
60     second->prev = head;  
61     second->next = third;  
62  
63     third->data = 30;  
64     third->prev = second;  
65     third->next = NULL;  
66  
67     printf("Before deletion: ");  
68     printlist(head);  
69  
70     deletenode(&head, second);  
71  
72     printf("After deletion: ");  
73     printlist(head);  
74  
75     return 0;
```

Test Input

stdin input (optional)

Output

```
Before deletion: 10 20 30  
After deletion: 10 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Prefix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int data)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = data;
16 }
17
18 int pop()
19 {
20     if (top == -1)
21     {
22         printf("Stack underflow\n");
23         return -1;
24     }
25     return stack[top--];
26 }
27
28 void display()
29 {
30     if (top == -1)
31     {
32         printf("Stack is empty\n");
33         return;
34     }
35     printf("Stack: ");
36
37     for (int i = top; i >= 0; i--)
38         printf("%d ", stack[i]);
39
40     printf("\n");
41 }
42
43 int main()
44 {
45     push(10);
46     push(20);
47     push(30);
48
49     display();
50
51     printf("Popped element: %d\n", pop());
52
53     display();
54
55     return 0;
56 }
```

Test Input

stdin input (optional)

Output

```
Stack: 30 20 10
Popped element: 30
Stack: 20 10
```

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Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notation - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int data)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15
16     stack[++top] = data;
17 }
18
19 int pop()
20 {
21     if (top == -1)
22     {
23         printf("Stack underflow\n");
24         return -1;
25     }
26
27     return stack[top--];
28 }
29
30 void display()
31 {
32     if (top == -1)
33     {
34         printf("Stack is empty\n");
35         return;
36     }
37
38     printf("Stack elements: ");
39
40     for (int i = top; i >= 0; i--)
41     {
42         printf("%d ", stack[i]);
43     }
44
45     printf("\n");
46 }
47
48 int main()
49 {
50     push(10);
51     push(20);
52     push(30);
53
54     display();
55
56     printf("popped element: %d\n", pop());
57
58     display();
59
60     return 0;
61 }
```

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Test Input

stdin input (optional)

Output

```
Stack elements: 30 20 10
popped element: 30
Stack elements: 20 10
```

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Recursion
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 void enqueue(int data)
9 {
10     // Check if queue is full
11     if ((rear + 1) % MAX == front)
12     {
13         printf("Queue full\n");
14         return;
15     }
16
17     // First element
18     if (front == -1)
19     {
20         front = rear = 0;
21     }
22     else
23     {
24         rear = (rear + 1) % MAX;
25     }
26
27     queue[rear] = data;
28 }
29
30 int dequeue()
31 {
32     if (front == -1)
33     {
34         printf("Queue empty\n");
35         return -1;
36     }
37
38     int data = queue[front];
39
40     if (front == rear)
41     {
42         front = rear = -1;
43     }
44     else
45     {
46         front = (front + 1) % MAX;
47     }
48
49     return data;
50 }
51
52 void display()
53 {
54     if (front == -1)
55     {
56         printf("Queue empty\n");
57         return;
58     }
59
60     int i = front;
61
62     printf("Queue: ");
63
64     while (1)
65     {
66         printf("%d ", queue[i]);
67
68         if (i == rear)
69             break;
70
71         i = (i + 1) % MAX;
72     }
73
74     printf("\n");
75 }
```

Test Input

stdin input (optional)

Output

```
Queue: 10 20 30 40 50
Queue: 30 40 50
```

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CO2, AT2, DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Prefix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Stack
5 marks**Q9**
Circular Queue
5 marks**Q10**
Stack
5 marks

10/10 answered

Q10 Stack

5 marks

Point out the error?

```
PEEK(stack);  
return stack[top+1];
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 void push(int data)  
9 {  
10     if (top == MAX - 1)  
11     {  
12         printf("Stack Overflow\n");  
13         return;  
14     }  
15     stack[++top] = data;  
16 }  
17  
18 int pop()  
19 {  
20     if (top == -1)  
21     {  
22         printf("Stack underflow\n");  
23         return -1;  
24     }  
25     return stack[top--];  
26 }  
27  
28 int peek()  
29 {  
30     if (top == -1)  
31     {  
32         printf("Stack is empty\n");  
33         return -1;  
34     }  
35     return stack[top];  
36 }  
37  
38 void display()  
39 {  
40     if (top == -1)  
41     {  
42         printf("Stack is empty\n");  
43         return;  
44     }  
45     printf("Stack elements: ");  
46     for (int i = top; i >= 0; i--)  
47     {  
48         printf("%d ", stack[i]);  
49     }  
50     printf("\n");  
51 }  
52  
53 int main()  
54 {  
55     push(10);  
56     push(20);  
57     push(30);  
58     display();  
59     printf("top element = %d\n", peek());  
60     printf("popped element = %d\n", pop());  
61     display();  
62     return 0;  
63 }
```

Submitted

Test Input

stdin input (optional)

Output

```
Stack elements: 30 20 10  
top element = 30  
popped element = 30  
Stack elements: 20 10
```

Pending manual review

